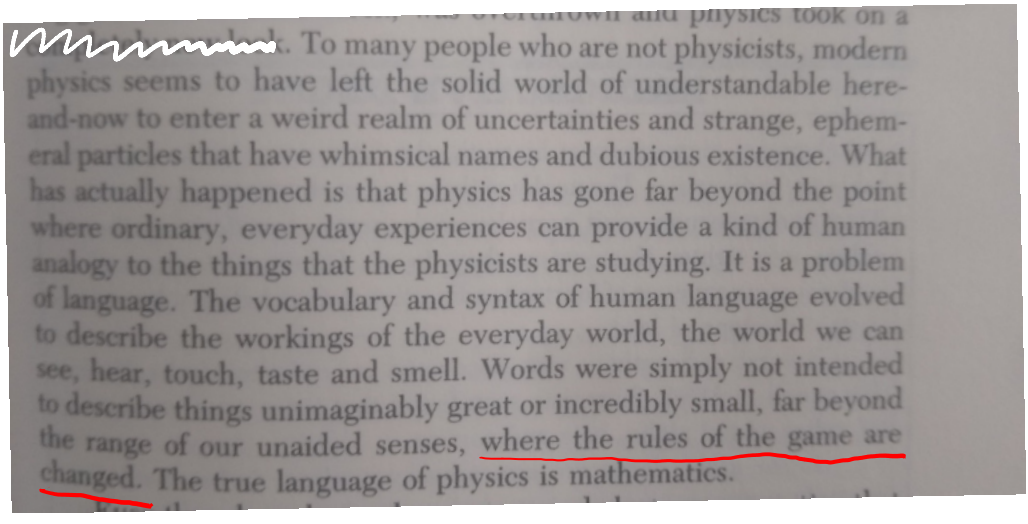
Goal: Give the biographical & scientific history of modern physics & see how it led to "QFT"

Physics - 1900

- Albert Michelson (of etherial fame)
  ~ "the great principles have been discovered, henceforth Physics will find truth in the 6th decimal."

Sheldon Glashow (Nobel 1979): written 1988



. To many people who are not physicists, modern physics seems to have left the solid world of understandable here-and-now to enter a weird realm of uncertainties and strange, ephemeral particles that have whimsical names and dubious existence. What has actually happened is that physics has gone far beyond the point where ordinary, everyday experiences can provide a kind of human analogy to the things that the physicists are studying. It is a problem of language. The vocabulary and syntax of human language evolved to describe the workings of the everyday world, the world we can see, hear, touch, taste and smell. Words were simply not intended to describe things unimaginably great or incredibly small, far beyond the range of our unaided senses, where the rules of the game are changed. The true language of physics is mathematics.

How did we go from Michelson to Glashow!?

2 Problems would emerge, each requiring a departure from our intuitional understanding of the world.

QM & Relativity

These form the bedrock of our understanding of fundamental Physics

To be clear, despite the odd implications & "Paradoxes" both are absolutely true about how our universe works

Trust only math & experiment, question your intuition!

Combining these, turns out, is Hard!
Has occupied Physicists from ~1920 – Present.
We didn't totally succeed, it's become a hugely broad subject.
All* modern Physics descends directly from these efforts.

The main* output of this effort was Quantum Field Theory
A framework for describing Particles & interactions that
- incorporates QM & Relativity.
  - A broad set of tools, perspectives, & methods that apply to a hugely broad set of theories. Very "general"
    - What Particles, what forces, etc.

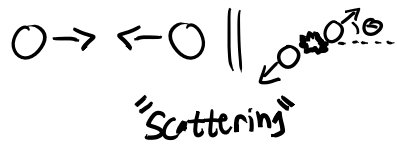
Timeline
Takeaways:

- Ebbs & flows of exp. vs. theory led (Post-Einstein)
- "What's fundamental" changes, sometimes rapidly!
- letting math & experiment lead the way, w/ many helping Players is how this huge, complex, Correct theory emerged!
- Why do we stop teaching at Bohr in 1913?
  So much more is now known!
  - see Glashow's quote.
    QFT is extremely complicated & lacks human intuition to help us. Nevertheless it works!
    - g-2 predicted & confirmed to 10 decimals
      best prediction in science!
    - Aside, a PSU Professor did a calculation suggesting a tension "at the 11th decimal", indicating a new force!
      Stay tuned! Cf. Michelson's quote. Aged like ironic wine

To be honest QFT is a few non-intuitive physical principles wrapped in decades of math, but I want to illustrate the gist of things.

How to QFT

Billiard ball picture:

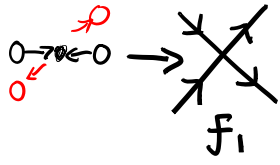


- QM says we want some Prob., $P(\theta)$
- Better yet an amplitude (just a (ℂ) number!) "the sqrt of probability" $P(\theta) = |f(\theta)|^2$
  - Why?  $a^2+b^2=c^2$

    but $\vec{a}, \vec{b}, \vec{c}$ are more useful.

    Similar idea, amplitudes are like vectors, & we care about $P = |f_1 + f_2 + \cdots|^2$

- Amplitudes are calculated from pictures called Feynman diagrams



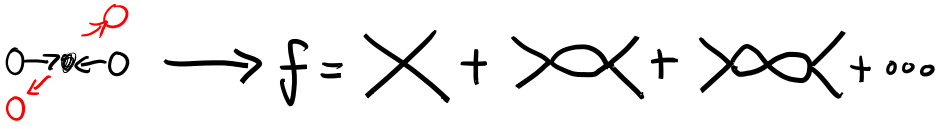
QFT give rules on how to turn a picture into an equation.

- Rel. says w/ enough energy, I can create mass (extra particles)
- QM says the prob. for this to happen gets smaller the longer the extra particles live $\Delta E \Delta t \geq \hbar$

So, the net effect is I need to add loops.



But it doesn't stop there!
QM says "if it can happen, it will"
So we draw all possible diagrams!

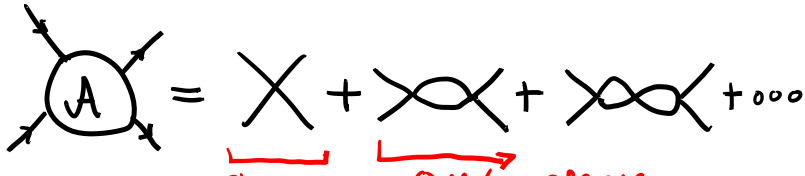


Each time you add lines you "make things more quantum",
(need more energy to create more particles, $\Delta E \Delta t \geq \hbar$)
So, we organize the list s.t. the things at the front are most likely



$a^2 \gg b^2 \Rightarrow c^2 \approx a^2$ (0°)

So, our picture is



Classic part (should agree w/ Newton) — QM/SR effects

Calculate as many as you can, w/ each correction getting smaller until you can't calculate one, or you declare good enuff

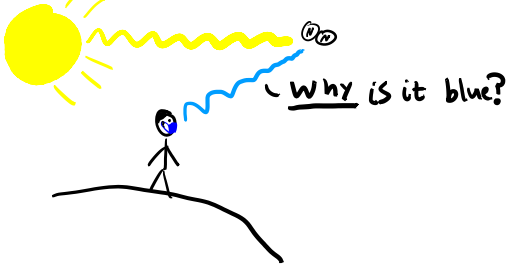
Nota bene: This technique not unique to QFT!
"neglect friction & air-resistance"

exact answer ~ idealized answer + corrections for friction/air resistance

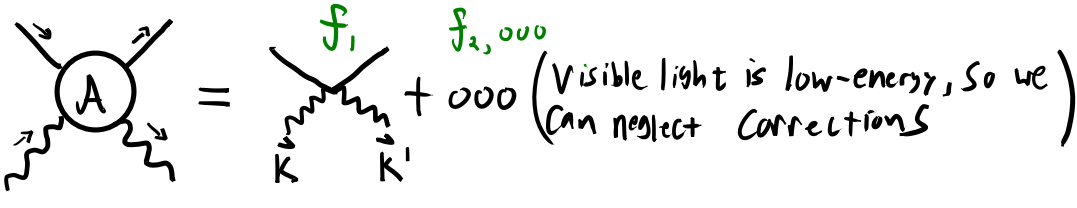
That's the basic idea of QFT!

Let's see an example.
Why is the sky blue? (Rayleigh, but why?)

Planck's constant, makes QM!



- incoming light, $K = h/\lambda$
- scattered light, $K' = h/\lambda'$
- scatters off N_2 molecules



(Visible light is low-energy, so we can neglect corrections)

Feynman Rules:

- Each cross $\sim g$ (A constant, maybe w/ units, says how strong a force is)
- Any vector (momentum) needs to be turned into a plain ol' #, since f is just a #.
- more, but that's all we need!

Note if $F_{light} < M_{N_2}$, then it's like throwing a ball at a brick wall, $\vec{k} = -\vec{k}'$, $|k| = |k'|$

$$\rightsquigarrow f \sim (g)(\vec{k} \cdot \vec{k}) \sim g k^2$$

So

$$P(\text{scatter}) = |f|^2 = g^2 k^4 = g^2 \left(\frac{h}{\lambda}\right)^4 \propto \frac{1}{\lambda^4}$$

Agrees w/ Rayleigh (1875), even though if he saw this he'd have no clue what we're doing!

Punchlines:

- Blue light is shorter wave length than most of the visible spectrum, so $P(\text{blue}) \gg P(\text{red})$
- Since more blue light scatters, the sky looks blue!
- At sunset the light travels through more of the atmosphere, so the blue light scatters so much it gets diffuse & the red light shows through. Also why pollution causes vivid sunsets, airborne particles will also scatter the light.

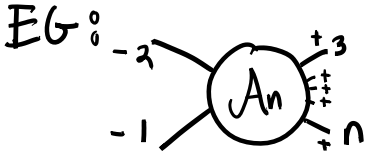
Also note! I never cared that N_2 is not fundamental, QFT, or it's cousin EFT, let you not care what is fundamental, as long as the energies involved aren't too big.
(eventually the light would break the bond between the Nitrogen & the situation changes a lot.)
This is a feature, not a bug!
Our "Best theory ever", the Standard Model has an impressive exp. track record, but if/when we discover more new particles, we can extend the theory naturally.

That wasn't so bad! Who says QFT is complicated?

Me. QFT is *designed* to make these calculations work naturally
The boogey man is the (tree) part we ignored.
Those parts *do* matter & the demands of QFT
led to entire fields of math being developed
just to calculate the first few diagrams.

My work is coming up w/ new tricks to
calculate these things. (or just understand them!)

EG:



n=4 ~ 3 diagrams
n=5 ~ 10 diagrams
n=6 ~ 38 diagrams
n=15 ~> 30 million diagrams!
Gets out of hand fast!

In particalar we focus on expressing things in a
way that makes manifest the *physical* principles.
Try to avoid

Using new techniques we know the answer for *any* n,
w/o having to calculate thousands of diagrams.

$$A_n(1^-,2^-,3^+,\dots,n^+) = \frac{\langle 12\rangle^4}{\langle 12\rangle\langle 23\rangle\langle 34\rangle\cdots\langle n1\rangle}$$

not just easier, an aesthetic eqn!